

## TWO-PARAMETER POWER SERIES DISTRIBUTIONS WITH APPLICATIONS

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*Key words:*

Birth-death model; generalized PSD with two parameters; maximum likelihood estimation; power series distribution (PSD); two-parameter geometric distribution; two-parameter logseries distribution; two-parameter Poisson distribution; size of free-forming groups.

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*Summary:*

Two-parameter power series distributions (PSD2) from the class described by the author (see South African Statist. J. (1980), 14, 1-15) are considered in more detail. It is shown how a birth-death model as well as models concerning sizes of free-forming groups give rise to these distributions, while the two-parameter Katz family can be extended to include the PSD2. Since PSD2 falls also into Patil's (see *Contributions to Statistics*, C.R. Rao (ed.), Pergamon Press, Oxford, 1964, pp. 335-344) class of generalized two-parameter PSD, maximum likelihood estimation follows on the same lines as given by Patil, but with the emphasis on the applications. The special cases of Poisson, logseries and geometric two-parameter distributions are considered with recurrence relations in probabilities and factorial moments. An example is given to show how the truncated two-parameter Poisson adequately fits data on sizes of free forming groups of people.

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### 1. INTRODUCTION

A class of multi-parameter power series distributions (PSD) generated from a PSD, was proposed by the author (see Steyn (1980)).

It was also shown how a two-parameter logseries distribution can be used to describe the distribution of sizes of free-forming groups of people. Apart from a good fit, no good motivation could then be given for a suitable underlying model.

In this article it is shown in section 2 how a birth-death model gives rise to the two-parameter PSD. It will first be shown how the Katz two-parameter family of distributions can be extended to include the class of generalized two-parameter PSD proposed by Patil (1964). The two-parameter PSD (Steyn (1980)) falls into the last mentioned class, and maximum likelihood estimation follows on the same lines as given by Patil (1964). This is shown in section 5.

In this paper an effort is made to derive a sound basis for the generalized two-parameter PSD and to derive further properties of these distributions.

Sections 3 and 4 give special cases of the two-parameter PSD under consideration together with their properties such as factorial moments and recurrence relations. Applications are given in section 6.

## 2. BIRTH-DEATH MODELS WITH LONG-TERM EQUILIBRIUM

Let the random variable  $X(t)$  denote the number of individuals in a population at time  $t$ . Let births occur at a rate  $\lambda(x,t)$  where  $x$  is the population size at the given time  $t$ . Similarly let deaths occur at the rate  $\mu(x,t)$ . Writing  $p(x,t)$  for  $P(X(t) = x)$ , it follows directly (see Boswell *et al.* (1979)) that for  $x = 1, 2, \dots$

$$(2.1) \quad \frac{\partial}{\partial t} p(x,t) = \lambda(x-1,t)p(x-1,t) - [\mu(x,t) + \lambda(x,t)]p(x,t) + \mu(x+1,t)p(x+1,t)$$

$$(2.2) \quad \frac{\partial}{\partial t} p(0,t) = -\lambda(0,t)p(0,t) + \mu(1,t)p(1,t),$$

where the possibility of multiple events in a small time-interval is excluded. Boswell *et al.* also showed that in the long term situation when  $\lambda(x,t)$  and  $\mu(x,t)$  tend to the limits  $\lambda(x)$  and  $\mu(x)$  as  $t$  tends to infinity, then, under appropriate conditions  $\frac{\partial}{\partial t} p(x,t) \rightarrow 0$  and the probabilities approach the equilibrium distributions  $p(x)$ , ( $x = 0, 1, 2, \dots$ ). From (2.1) and (2.2) it follows that

$$(2.3) \quad p(x+1)/p(x) = \lambda(x)/\mu(x+1), \quad x = 0, 1, 2, \dots$$

These probability ratios together with the constraint that the sum of probabilities is one, describe the entire behaviour of the equilibrium process  $X(t)$ .

Using the principles of these birth-death models, the size of free-forming groups can also be described by (2.3), with  $\lambda(x)h$  being the probability that an individual joins a group of size  $x$  in a time interval  $h$ . The probability of departure is likewise  $\mu(x)h$ . Since groups of size zero are meaningless, (2.3) is then defined for  $x = 1, 2, \dots$

The linear one-step transition (LOST) models of Cohen (1971), when  $\lambda(x)$  and  $\mu(x)$  are linear functions of  $x$ , is a special case of the birth-death models given by (2.3).

The truncated Poisson model (see Coleman (1964), Cohen (1971)) follows from (2.3) when  $\lambda(x)$  is a positive constant and  $\mu(x) = x$ . Cohen also described the truncated negative binomial where  $\lambda(x) = a + bx$  and  $\mu(x) = x$ . Both these models give rise to distributions which belong to the Katz two-parameter family (see Johnson and Kotz, 1969, p. 37) of discrete distributions.

Instead of the Katz family where  $\mu(x) = x$  and  $\lambda(x) = a + bx$ , with  $a > 0$ ,  $b < 1$  and  $x = 0, 1, \dots$ , a more general family will be considered. Take  $\mu(x) = x$ , as before, but let

$$(2.4) \quad \lambda(x) = \lambda(x; \theta, \delta),$$

where  $\lambda(x; \theta, \delta)$  is some function of  $x$  depending on the two parameters  $\theta$  and  $\delta$ . In terms of sizes of free-forming groups, this means that the probability that an individual joins a group of size  $x$  is proportional to  $\lambda(x; \theta, \delta)$  and the probability of departure is proportional to the size of the group from which the individual departs.

In the following section it will be shown that the so-called generalized power series distributions with two parameters (see Patil (1964)) falls into the class of distributions described by (2.3) and (2.4).

### 3. THE TWO-PARAMETER POWER SERIES DISTRIBUTION

First consider a power series distribution (PSD) with parameter  $\theta (> 0)$ :

$$(3.1) \quad p(x) = a_x \theta^x / f(\theta), \quad x = 0, 1, 2, \dots$$

with series function

$$(3.2) \quad f(\theta) = \sum_{x=0}^{\infty} a_x \theta^x,$$

$f(\theta)$  being finite and differentiable and  $a_x$  numbers depending on  $x$ .

From (3.2) it follows that for  $k = 0, 1, 2, \dots$

$$f^{(k)}(\theta) = \frac{d^k}{d\theta^k} f(\theta) = \sum_{x=k}^{\infty} x(x-1)\dots(x-k+1) a_x \theta^{x-k}$$

and therefore

$$(3.3) \quad f^{(k)}(0) = k! a_k$$

The probability ratio from (3.1) and (3.3) is

$$p(x+1)/p(x) = \theta f^{(x+1)}(0) / \{(x+1)f^{(x)}(0)\},$$

which implies that

$$(3.4) \quad \lambda(x) = \theta f^{(x+1)}(0) / f^{(x)}(0) = \lambda(x; \theta, 0).$$

Patil (1964) generalized the one-parameter PSD to a two-parameter distribution (GPSD2) as follows:

$$(3.5) \quad p(x) = a_x(\delta) \theta^x / g(\theta, \delta), \quad x \in T$$

where  $T$  is a non-null countable subset (with no limit point) of real numbers,

$$(3.6) \quad g(\theta, \delta) = \sum_{x \in T} a_x(\delta) \theta^x,$$

$a_x(\delta) > 0$  depending also on the parameter  $\delta$ , while  $\theta \geq 0$ . Again  $g(\theta, \delta)$  is assumed to be finite and differentiable w.r.t.  $\theta$ . Taking  $T = \{0, 1, 2, \dots\}$  it is clear that for the GPSD2

$$(3.7) \quad \lambda(x) = \theta D_{\theta}^{x+1} g(0, \delta) / D_{\theta}^x g(0, \delta) = \lambda(x, \theta, \delta), \quad \text{where } D_{\theta}^x g \text{ denotes } \partial^x g / \partial \theta^x.$$

The generalized form of a PSD given by Steyn (1980) where additional parameters are involved, is defined by the probability generating function,

$$(3.8) \quad P(t) = f(\theta_1 t + \theta_2 t^2) / f(\theta_1 + \theta_2),$$

with  $f$  defined in (3.2) and  $\theta_1 > 0, \theta_2 > 0$ .

Using the fact that for  $h(t) = \theta_1 t + \theta_2 t^2$ ,  $h(0) = 0$ ,  $h^{(1)}(0) = \theta_1$ ,  $h^{(2)}(0) = 2\theta_2$  and  $h^{(r)}(0) = 0$  for all  $r \geq 3$ , it follows that for  $x = 0, 1, 2, \dots$

$$\begin{aligned}
 (3.9) \quad p(x) &= \left( \frac{\partial^x}{\partial t^x} P(t) \Big|_{t=0} \right) / x! \\
 &= \sum_{r=0}^{[\frac{1}{2}x]} \frac{f^{(x-r)}(0)}{(x-2r)!r!} \theta^{x-2r} \theta_2^r / f(\theta_1 + \theta_2).
 \end{aligned}$$

([a] is the largest integer  $\leq a$ .)

Reparameterization of  $\theta_1$  and  $\theta_2$  through  $\theta = \theta_1$  and  $\delta = \theta_2/\theta_1^2$ , yields a special case of Patil's GPSD2 as defined in (3.5) with  $g(\theta, \delta) = f(\theta + \delta\theta^2)$ ,  $\theta > 0$  and

$$(3.10) \quad a_x(\delta) = \sum_{r=0}^{[\frac{1}{2}x]} \frac{f^{(x-r)}(0)\delta^r}{(x-2r)!r!}, \quad x = 0, 1, 2, \dots$$

Note that the PSD distributions are special cases for  $\delta = 0$  of this two-parameter PSD. When  $\delta \neq 0$  the notation PSD2 will be used. Truncation of  $p(x)$  at zero only involves substitution of  $f(\theta + \delta\theta^2)$  by  $f(\theta + \delta\theta^2) - f(0)$ .

### Special cases of PSD2-distributions

(i) The two-parameter Poisson distribution:

For  $f(t) = e^t$ , so that  $f(\theta z + \delta\theta^2 z^2) = e^{\theta z + \delta\theta^2 z^2}$  and from the fact that  $f(0) = 1 = f^{(k)}(0)$  for  $k = 1, 2, \dots$ , it follows that

$$(3.11) \quad p(x) = \theta^x \exp(-\theta - \delta\theta^2) \sum_{r=0}^{[\frac{1}{2}x]} \frac{\delta^r}{(x-2r)!r!}, \quad x = 0, 1, 2, \dots$$

This is the Hermite distribution described by Kemp and Kemp (1965) from which also follows that

$$p(x) = \exp(-\theta - \delta\theta^2) (\theta^x / x!) (2\delta)^{\frac{1}{2}x} H_n^*((2\delta)^{-\frac{1}{2}}),$$

where  $-H_n^*(x)$  is the modified Hermite polynomials as defined by Kemp and Kemp (1965).

(ii) The two-parameter logseries distribution:

For  $f(t) = -\ln(1-t)$  it follows that  $f(0) = 0$ ,  $f^{(k)}(0) = (k-1)!$ ,  $k = 1, 2, \dots$ . Therefore for  $x = 1, 2, \dots$

$$(3.12) \quad p(x) = -\theta^x \sum_{r=0}^{[\frac{1}{2}x]} \frac{(x-r-1)! \delta^r}{(x-2r)! r!} / \ln(1-\theta-\delta\theta^2).$$

Note that the coefficient of  $-\theta^x/(x \ln(1-\theta-\delta\theta^2))$  is the hypergeometric function  $F(-\frac{1}{2}x; -\frac{1}{2}(x-1); -x; -4\delta)$  as defined by Abramowitz and Stegun (1970, chapter 15).

(iii) The two-parameter geometric distribution:

For  $f(t) = 1/(1-t)$ ,  $f(0) = 1$ ,  $f^{(k)}(0) = k!$  for  $k = 1, 2, \dots$

$$(3.13) \quad p(x) = \theta^x (1-\theta-\delta\theta^2) \sum_{r=0}^{[\frac{1}{2}x]} \binom{x-r}{r} \delta^r \\ = \theta^x (1-\theta-\delta\theta^2) F(-\frac{1}{2}x; -\frac{1}{2}(x-1); -x; -4\delta).$$

Apart from the genesis of the PSD2 described by Steyn (1980), an informative way of obtaining these distributions follows therefore from the birth-death models (or free-forming group models). In order to apply the PSD2 in practice, recursion formulas for derivatives of  $a_x(\delta)$ , probabilities  $p(x)$  and factorial moments are given in the following section.

## 4. RECURSION FORMULAS FOR GENERAL AND SPECIAL CASES

First consider

$$\begin{aligned}
 (4.1) \quad \frac{d}{d\delta} a_x(\delta) &= \sum_{r=1}^{[\frac{1}{2}x]} r \frac{f(x-r)(0)}{(x-2r)!r!} \delta^{r-1}, \quad \text{for } x = 2, 3, 4, \dots \\
 &= \sum_{s=0}^{[\frac{1}{2}x]-1} \frac{f(x-s-1)(0)}{(x-2(s+1))!s!} \delta^s \\
 &= \sum_{s=0}^{[\frac{1}{2}(x-2)]} \frac{f(x-2-s)(0)}{(x-2-2s)!s!} \delta^s \left\{ \frac{f(x-1-s)(0)}{f(x-2-s)(0)} \right\}
 \end{aligned}$$

From (3.10) it follows that both

$$(4.2) \quad a_0(\delta) = f(0), \quad a_1(\delta) = f^{(1)}(0)$$

which do not depend on  $\delta$  and therefore

$$(4.3) \quad \frac{d}{d\delta} a_x(\delta) = 0$$

for  $x = 0$  and  $1$ . In the same way as (4.1) for  $x = 4, 5, 6, \dots$

$$(4.4) \quad \frac{d^2}{d\delta^2} a_x(\delta) = \sum_{s=0}^{[\frac{1}{2}(x-4)]} \frac{f(x-4-s)(0)}{(x-4-2s)!s!} \delta^s \frac{f(x-2-s)(0)}{f(x-4-s)(0)}$$

For appropriate choices of  $f(t)$  equations (4.1) and (4.4) lead to recurrence relations. For the special cases considered in the previous section, it follows that:

(i) Two-parameter Poisson distribution:

$$\begin{aligned}
 (4.5) \quad \frac{d}{d\delta} a_x(\delta) &= a_{x-2}(\delta), \quad x = 2, 3, \dots \\
 \frac{d^2}{d\delta^2} a_x(\delta) &= a_{x-4}(\delta), \quad x = 4, 5, \dots
 \end{aligned}$$

(ii) Two-parameter logseries distribution:

$$\begin{aligned}
 (4.6) \quad \frac{d}{d\delta} a_x(\delta) &= (x-2)a_{x-2}(\delta) - \delta \frac{d}{d\delta} a_{x-2}(\delta), \quad x = 2, 3, \dots \\
 \frac{d^2}{d\delta^2} a_x(\delta) &= (x-3)(x-4)a_{x-4}(\delta) - 2(x-3)\delta \frac{d}{d\delta} a_{x-4}(\delta) \\
 &\quad + \delta^2 \frac{d^2}{d\delta^2} a_{x-4}(\delta), \quad x = 4, 5, \dots
 \end{aligned}$$

(iii) Two-parameter geometric distribution:

$$\begin{aligned}
 (4.7) \quad \frac{d}{d\delta} a_x(\delta) &= (x-1)a_{x-2}(\delta) - \delta \frac{d}{d\delta} a_{x-2}(\delta), \quad x = 2, 3, \dots \\
 \frac{d^2}{d\delta^2} a_x(\delta) &= (x-2)(x-3)a_{x-4}(\delta) - 2(x-1)\delta \frac{d}{d\delta} a_{x-4}(\delta) \\
 &\quad + \delta^2 \frac{d^2}{d\delta^2} a_{x-4}(\delta), \quad x = 4, 5, \dots
 \end{aligned}$$

Using (3.5) it follows for GPSD2 and therefore also for PSD2 that

$$(4.8) \quad p(x+1) = \theta(a_{x+1}(\delta)/a_x(\delta))p(x), \quad \text{for } x = 0, 1, 2, \dots,$$

for which follows  $p(x) = \theta^{x-1}a_x(\delta)p(1)$  or  $p(x) = \theta^x a_x(\delta)p(0)$ .

Noting that

$$f(\theta + \delta\theta^2) = \sum_{x=0}^{\infty} a_x(\delta)\theta^x,$$

it follows that

$$\frac{\partial^r}{\partial \theta^r} f(\theta + \delta\theta^2) = \sum_{x=r}^{\infty} x(x-1) \dots (x-r+1)a_x(\delta)\theta^{x-r}.$$

It is apparent that the  $r$ -th factorial moment of the PSD2 is given by

$$(4.9) \quad \mu(r) = \theta r f(r) (\theta + \delta \theta^2) / f(\theta + \delta \theta^2).$$

Therefore, writing  $f$  for  $f(\theta + \delta \theta^2)$ :

$$\frac{\partial}{\partial \theta} \mu(r) = r \theta^{r-1} f(r) / f + \theta^r \{ (f(r+1)/f) - f(r) f'(1)/f^2 \} \cdot (1 + 2\delta \theta),$$

it follows that

$$(4.10) \quad \mu(r+1) = \theta \frac{\partial}{\partial \theta} \mu(r) + \mu(r) \{ (1 + 2\delta \theta) \mu(1) - r \}.$$

## 5. MAXIMUM LIKELIHOOD ESTIMATION OF $\theta$ AND $\delta$

Writing the PSD2 as a GPSD2, it follows from Patil (1964) that the maximum likelihood estimators (m.l.e.) for  $\theta$  and  $\delta$  can be obtained from the solution of the equations (based on a random sample  $x_1, x_2, \dots, x_n$ ):

$$(5.1) \quad \bar{x} = \mu(\theta, \delta)$$

$$(5.2) \quad \frac{\partial}{\partial \delta} \ln f(\theta + \delta \theta^2) = \frac{n}{\sum_{i=1}^n} \frac{d}{d\delta} \ln a_{x_i}(\delta) / n,$$

where  $\mu(\theta, \delta)$  is the mean of a PSD2, which is according to Steyn (1980):

$$(5.3) \quad \mu(\theta, \delta) = (1 + \frac{\delta \theta}{1 + \delta \theta}) \mu^*(\theta + \delta \theta^2),$$

with  $\mu^*(\lambda)$  the mean of a PSD with series function  $f(\lambda)$ .

From (4.9) it follows that  $\mu(\theta, \delta)$  can also be written as

$$(5.4) \quad \mu(\theta, \delta) = \theta \frac{\partial}{\partial \theta} f(\theta + \delta \theta^2) / f(\theta + \delta \theta^2)$$

and consequently (5.2) is simplified by using (5.1), (5.4) and (4.3):

$$(5.5) \quad \frac{\frac{\partial}{\partial \delta} f(\theta + \delta \theta^2)}{\theta \frac{\partial}{\partial \theta} f(\theta + \delta \theta^2)} = \frac{1}{n\bar{x}} \sum_{r=2}^k \frac{n_r \frac{d}{d\delta} a_r(\delta)}{a_r(\delta)},$$

where the values  $n_1, n_2, \dots, n_k$  are the frequencies of  $1, 2, \dots, k$  as obtained in the sample.

The left-hand side of (5.5) reduces to  $\theta/(1+2\delta\theta)$  and therefore  $\theta$  can be written in terms of  $\delta$  as follows:

$$(5.6) \quad \theta = \frac{\sum_{r=2}^k n_r \frac{d}{d\delta} a_r(\delta) / a_r(\delta)}{n\bar{x} - 2\delta \sum_{r=2}^k n_r \frac{d}{d\delta} a_r(\delta) / a_r(\delta)}.$$

Writing  $\mu(\theta(\delta), \delta)$  for  $\mu(\theta, \delta)$  when  $\theta$  is substituted by (5.6) and defining

$$H(\delta) = \mu(\theta(\delta), \delta) - \bar{x}$$

from (5.1),  $\delta$  may be obtained iteratively through application of the Newton-Raphson method:

$$(5.7) \quad \delta^{(r)} = \delta^{(r-1)} - H(\delta^{(r-1)}) / \frac{d}{d\delta} H(\delta^{(r-1)}), \quad r = 1, 2, \dots$$

Here  $\delta^{(0)}$  is an initial value for  $\delta$  and  $\delta^{(r)}$  are the  $\delta$ -values obtained iteratively, while  $\hat{\delta}$  is the limiting value of  $\delta^{(r)}$ , which may be taken to be the m.l.e. for  $\delta$  with  $\hat{\theta}$  the m.l.e. for  $\theta$  obtained from  $\hat{\delta}$  in (5.6).

$$(5.8) \quad \begin{aligned} \frac{d}{d\delta} H(\delta) &= \frac{d}{d\delta} \left[ \frac{1+2\delta\theta(\delta)}{1+\delta\theta(\delta)} \mu^*(\theta(\delta) + \delta\theta^2(\delta)) \right] \\ &= \frac{\theta(\delta) + \delta\theta'(\delta)}{(1+\delta\theta(\delta))^2} \mu^*(\theta(\delta) + \delta\theta^2(\delta)) \\ &\quad + \frac{1+2\delta\theta(\delta)}{1+\delta\theta(\delta)} \theta(\delta)(\theta(\delta) + 2\delta\theta'(\delta)) \left( \frac{d}{dt} \mu^*(t) \right) \Big|_{t=\theta(\delta)+\delta\theta^2(\delta)} \end{aligned}$$

with

$$(5.9) \quad \theta'(\delta) = \frac{c_2(\delta)}{n\bar{x} - 2\delta c_1(\delta)} + \frac{2c_1(\delta)}{(n\bar{x} - 2\delta c_1(\delta))^2} \cdot (\delta c_2(\delta) - c_1(\delta)),$$

where

$$c_1(\delta) = \sum_{r=1}^k n_r \frac{d}{d\delta} a_r(\delta) / a_r(\delta), \quad \text{and}$$

$$c_2(\delta) = \sum_{r=2}^k n_r \frac{d^2}{d\delta^2} \ln a_r(\delta) = \sum_{r=2}^k n_r \left\{ \frac{a_r(\delta) \frac{d^2}{d\delta^2} a_r(\delta) - \left( \frac{d}{d\delta} a_r(\delta) \right)^2}{a_r^2(\delta)} \right\}$$

If difficulty is experienced with divergence of  $\delta^{(r)}$ -values in (5.7) a more appropriate initial guess for  $\delta^{(0)}$  has to be made.

From (4.8) it follows that

$$(5.10) \quad \begin{aligned} p(1) &= \theta(a_1(\delta)/a_0(\delta))p(0) \quad \text{and} \\ p(2) &= \theta(a_2(\delta)/a_1(\delta))p(1), \end{aligned}$$

and from (3.10) the explicit expressions

$$(5.11) \quad \theta = n_1 f(0) / (n_0 f^{(1)}(0)) \quad \text{and}$$

$$(5.12) \quad \delta = n_2 n_0 (f^{(1)}(0))^2 / (n_1^2 f(0)) - \frac{1}{2} f^{(2)}(0) / f^{(1)}(0) \quad \text{follow,}$$

which are obtained by using  $n_1/n_0$  and  $n_2/n_1$  as estimates for  $p(1)/p(0)$  and  $p(2)/p(1)$ . For initial  $\delta^{(0)}$ -value, (5.12) can be used. In cases where  $p(0) = 0$  (e.g. logseries case),  $p(3) = \theta(a_3(\delta)/a_2(\delta))p(2)$  should be used in (5.10), from which it follows that

$$(5.13) \quad \delta^2 \frac{n_3}{n_2} f^{(1)}(0) + \delta \left\{ \left( \frac{n_3}{2n_2} - \frac{n_2}{n_1} \right) f^{(2)}(0) + \frac{n_3}{2n_2} f^{(3)}(0) \right\} \\ + f^{(3)}(0) \left( \frac{n_3}{4n_2} f^{(2)}(0) / f^{(1)}(0) - \frac{1}{6} \frac{n_2}{n_1} \right) = 0.$$

The appropriate root of this quadratic expression serves as an initial  $\delta$ -value.

For the special cases considered in the previous sections  $\mu^*(t)$  and  $\frac{d}{dt}\mu^*(t)$  are:

$$(5.14) \quad \mu^*(t) = \begin{cases} t & \text{for Poisson} \\ -t/t \ln(1-t) & \text{for logseries} \\ t/(1-t) & \text{for geometric} \end{cases}$$

$$(5.15) \quad \frac{d}{dt}\mu^*(t) = \begin{cases} 1 & \text{for Poisson} \\ 1/\{(1-t)[\ln(1-t)]^2\} & \text{for logseries} \\ 1/(1-t)^2 & \text{for geometric} \end{cases}$$

The information matrix is given by

$$S = \begin{pmatrix} I_{11} & I_{12} \\ I_{12} & I_{22} \end{pmatrix}, \quad \text{with}$$

$$I_{11} = n \frac{\partial \mu(\theta, \delta)}{\partial \theta} / \theta, \quad I_{12} = n \frac{\partial \mu(\theta, \delta)}{\partial \delta} / \theta$$

and

$$I_{22} = n \left[ \frac{\partial^2}{\partial \delta^2} \ln f(\theta + \delta \theta^2) - \frac{\partial^2}{\partial \delta^2} f(\theta + \delta \theta^2) / f(\theta + \delta \theta^2) + E \left\{ \frac{d}{d\delta} \ln a_X(\delta) \right\}^2 \right].$$

From a large sample the following estimate can be used for the last term of  $I_{22}$ :

$$\frac{1}{n} \sum_{r=2}^k n_r \left( \frac{d}{d\delta} \ln a_r(\delta) \right) = c_1(\delta) / n.$$

By using (5.4) and (4.9) it is convenient to write

$$I_{11} = \frac{n(1+2\delta\theta)}{\theta^2} (\mu(2) - \mu(1)),$$

and also from (5.3):

$$I_{12} = n \left\{ \mu^*(\theta + \delta\theta^2) / (1 + \delta\theta)^2 + \frac{\theta(1 + 2\delta\theta)}{1 + \delta\theta} \left( \frac{d}{dt} \mu^*(t) \right) \Big|_{t=\theta + \delta\theta^2} \right\}.$$

Inversion of matrix  $S$  above supplies the asymptotic covariance matrix of the m.l.e. for  $\theta$  and  $\delta$ . Substitution of  $\hat{\theta}$  and  $\hat{\delta}$  for  $\theta$  and  $\delta$  gives the estimated asymptotic variances and covariances.

## 6. AN EXAMPLE

Consider example 2 in Steyn (1980) where the size of free-forming groups of people were observed at the Garlicks restaurant, Pretoria. The results are given in Table 1.

TABLE 1

| Group size<br>(r) | Observed<br>frequency<br>( $n_r$ ) | Expected frequencies of<br>two-parameter power series |                        |
|-------------------|------------------------------------|-------------------------------------------------------|------------------------|
|                   |                                    | Logseries                                             | Poisson<br>(truncated) |
| 1                 | 136                                | 140                                                   | 138                    |
| 2                 | 123                                | 122                                                   | 120                    |
| 3                 | 27                                 | 21                                                    | 26                     |
| 4                 | 9                                  | 11                                                    | 12                     |
| 5                 | 2                                  | 3                                                     | 2                      |
| 6                 | 1                                  | 2                                                     | 1                      |
| 7+                | 1                                  | 0                                                     | 0                      |
|                   | 299                                | 299                                                   | 299                    |

$$\hat{\theta} = 0,175 \quad \hat{\theta} = 0,236$$

$$\hat{\delta} = 4,473 \quad \hat{\delta} = 3,209$$

$$\chi^2 = 2,4 \quad \chi^2 = 0,409$$

$$(0,25 < P < 0,5) \quad (0,75 < P < 0,9)$$

In comparison to the best fit which was obtained before namely by using the two-parameter logseries distribution, the truncated two-parameter Poisson gives even a better fit (using  $\chi^2$  as criterion). Where the ordinary truncated Poisson failed to fit the data (see Steyn (1980)), it seems natural that the extension to a distribution with an extra parameter gives rise to a better fit.

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